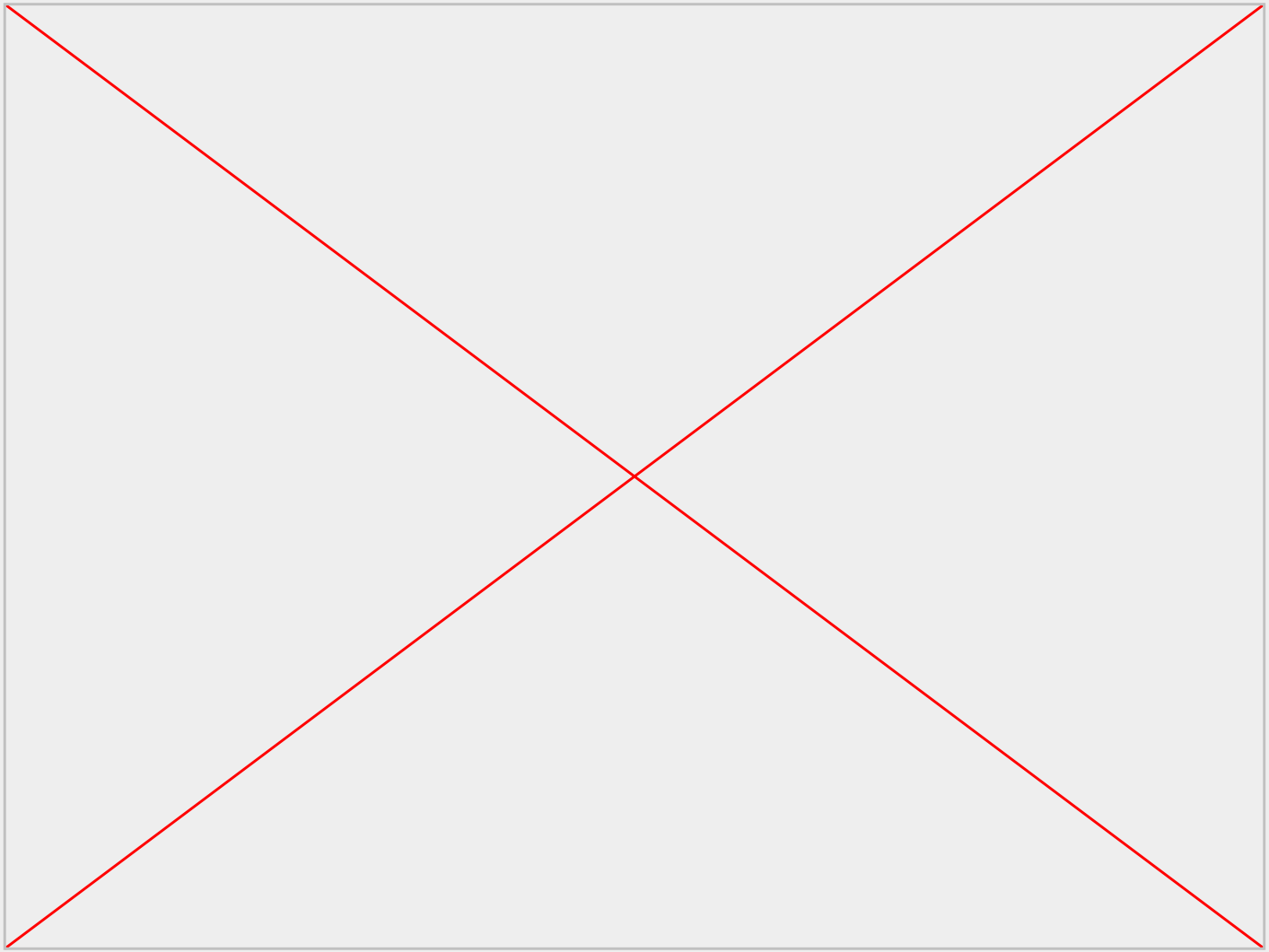
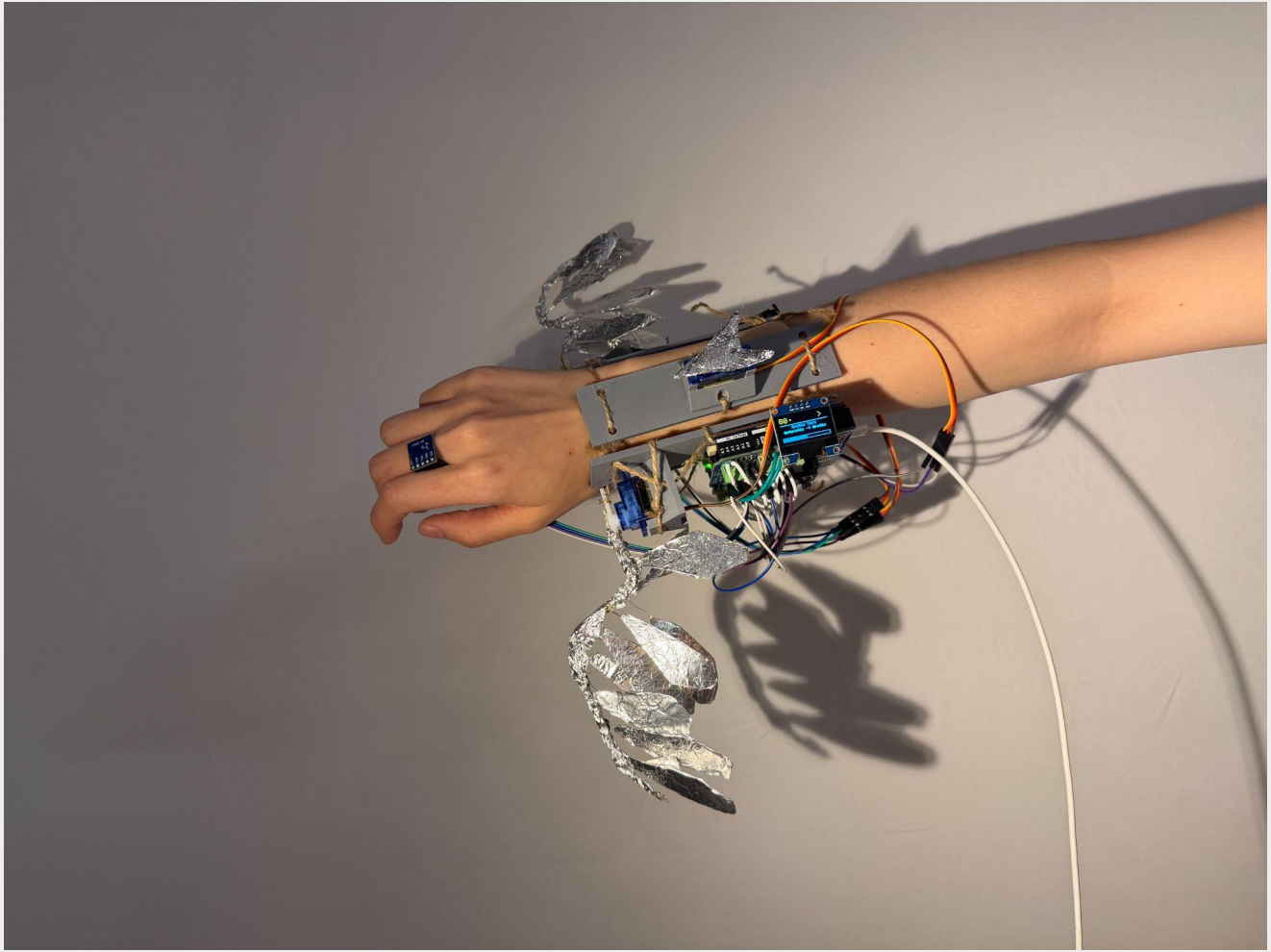
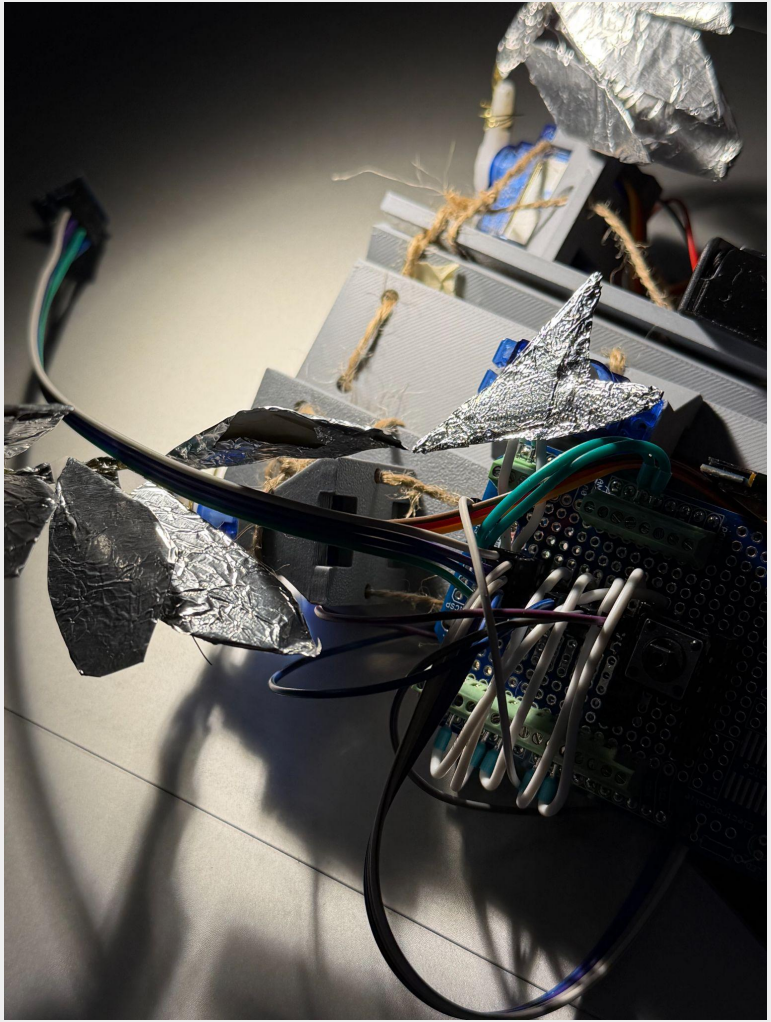
Project 1 Documentation

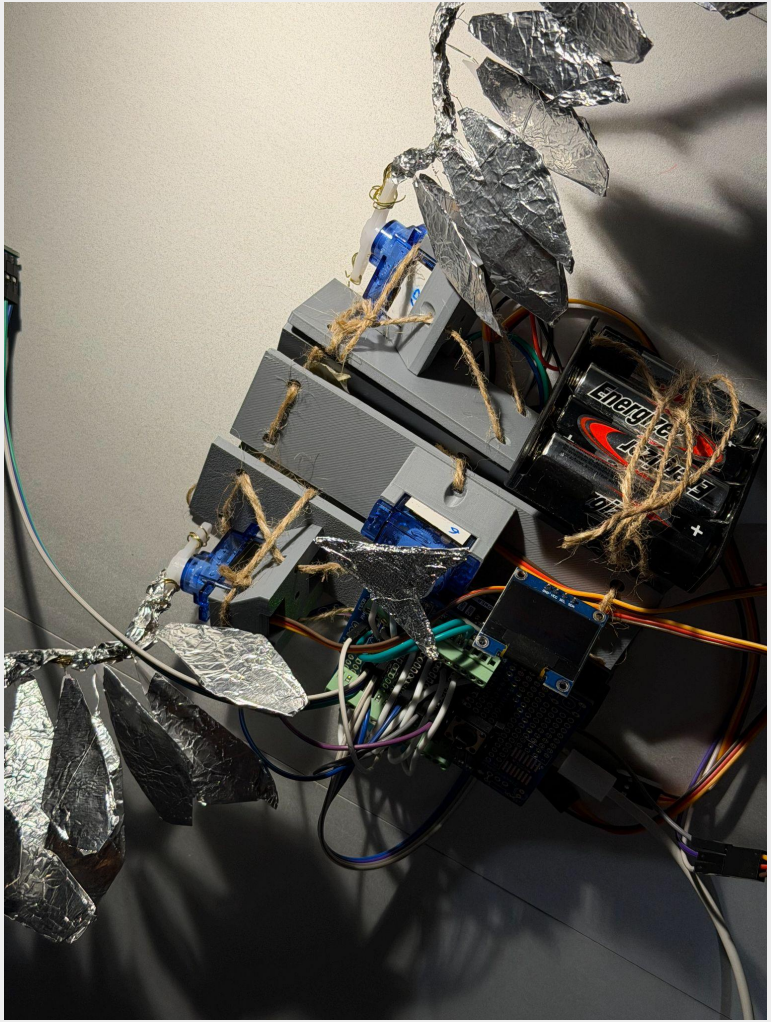
Julianne Jin

Final









Project statement

This project explores the integration of a bird's electromagnetic perception onto a human arm, with an output wing feature that responds to the arm's orientation.

Initially, I intended to create a highly realistic wing motion that could fold, expand, and move both up and down, mirroring a real bird's flight. However, due to limitations in servo power and the complexities of the physical modeling, I streamlined the design to focus only on a vertical flapping motion.

The construction of the wing/feathers utilized lightweight materials, including metal wire and aluminum foil. The system incorporates three servos, a battery pack, and a five-panel 3D-printed arm cuff. To minimize visual clutter and reduce wiring, I opted to solder the circuit board. Grass rope was chosen for tying the units together, facilitating easy assembly and modification.

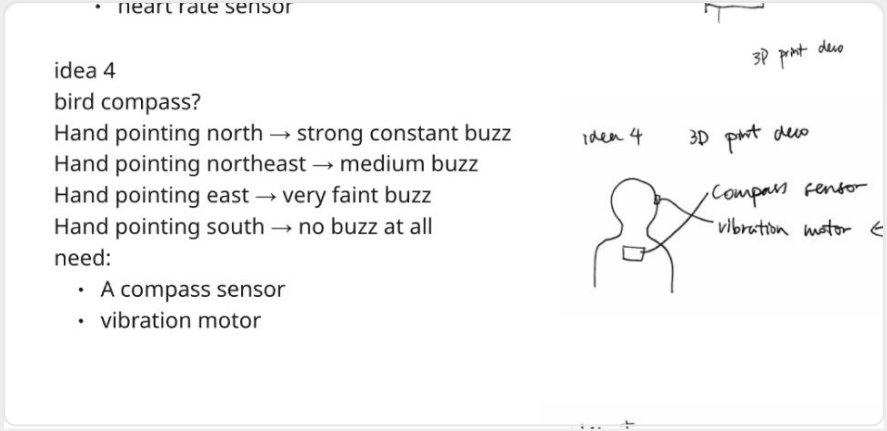
This work prompts reflection on how our biological constraints, our inability to perceive magnetic fields, infrasound, or ultraviolet light, shape our understanding of the world. These are not intellectual shortcomings but hardware limitations. What we cannot perceive, we cannot respond to, which inherently defines the boundaries of our reality. The neuroplasticity of the human brain suggests an ability to adapt to new sensory input. This leads to the central question: what transformation occurs when the body internalizes a new sense over an extended period of time?

Project logic

1. Start circuit
2. Display: current bird type out of total of 5, switch upon button
3. Setup compass, read compass direction and map direction to desired bird migration direction
4. If aligned → wing move in sine wave in larger range
5. If not aligned → wing move in smaller range and speed

To optimize the wing speed and efficiency, added smoothing effect for compass data to be translated into wing speed.

Initial Idea



The Bird Compass Concept

Birds navigate using an internal magnetic compass—a sense called magnetoreception—primarily by detecting the inclination of Earth's magnetic field lines through specialized proteins in their eyes.

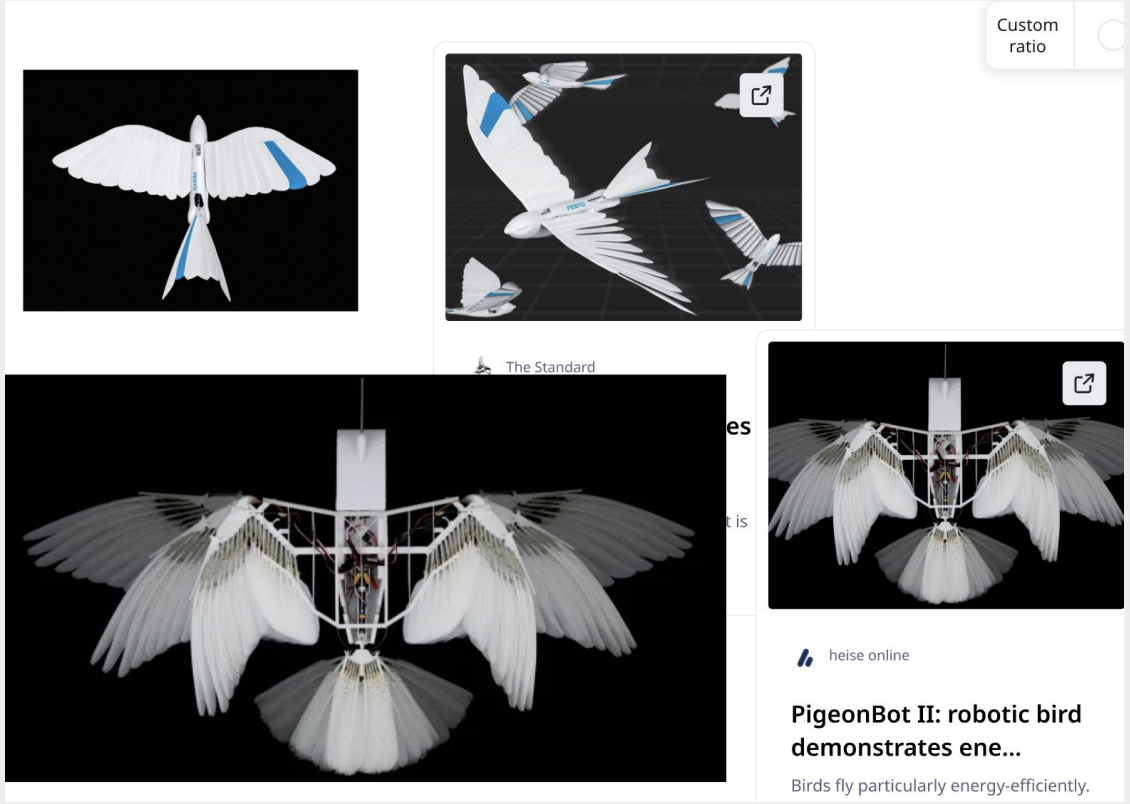
This project explores giving the human body a similar sense. Artificial electromagnetic signals (Wi-Fi, electronics) can interfere with this perception. By translating these invisible fields into physical touch, we can expand our sensory boundaries.

Haptic Feedback Mapping:

- **North:** Strong constant buzz
- **Northeast:** Medium buzz
- **East:** Very faint buzz
- **South:** No buzz

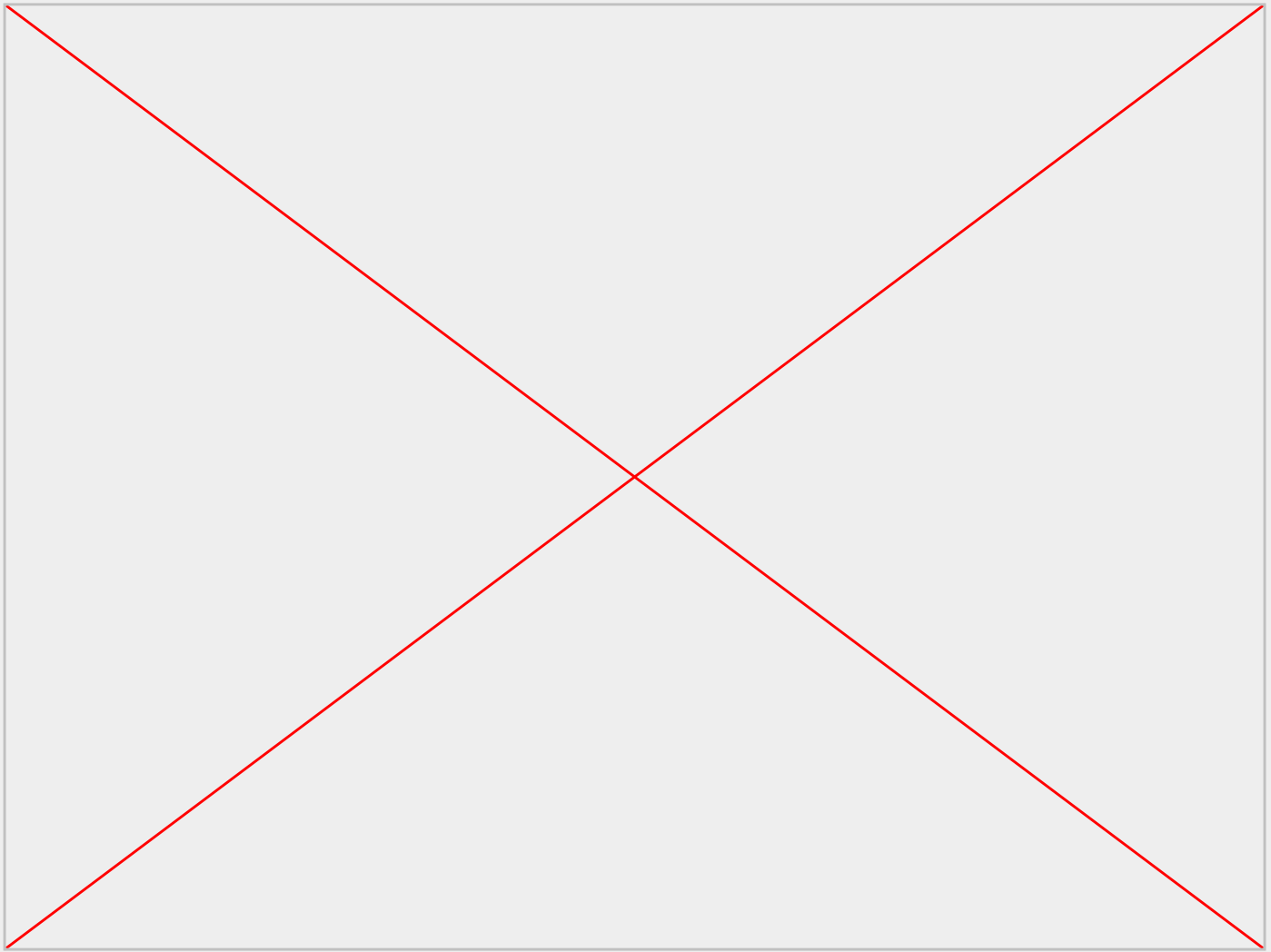
Hardware: Compass sensor, vibration motor, 3D printed arm cuff.

Reference


Custom ratio
The Standard
es
t is
heise online
PigeonBot II: robotic bird demonstrates ene...
Birds fly particularly energy-efficiently.

Project 2 documentation

Final



Final



Concept Statement

This project explores the experience of standing in front of a window. Windows often appear in poems and images as thresholds between inside and outside, private thought and the changing world. They frame weather, time of day, drifting attention, and mood. Rather than treating a window as only an architectural opening, I see it as a surface for projection, reflection, and subconscious atmosphere.

The work also reflects my childhood imagination and memory of looking out through windows. Growing up, windows carried a sense of distance, curiosity, and possibility — a connection to a larger life outside the room. The green tones and filtered light reference my personal association between windows, nature, and an imagined “greener” world beyond immediate space. Rather than representing a literal landscape, the piece reconstructs the emotional feeling of looking outward: anticipation, quiet reflection, and the sense that something exists beyond the boundary of the wall.

I created a fake window that reacts to both the viewer and the surrounding environment. When placed on a wall, it acts like a prosthetic opening, making the room feel more breathable and alive. Instead of fully simulating a naturalistic view, such as blue sky or realistic sunlight, the piece uses green and ochre light, leaf-patterned muslin, and translucent bioplastic to exaggerate my personal memory of what a window feels like. It does not recreate nature directly; it reconstructs the sensation of filtered light, seasonal color, and quiet interior reflection.

The window is built from three main systems: an Arduino circuit, an LED light panel, and a bioplastic window surface. The Arduino uses a photoresistor and ultrasonic sensor to respond to room brightness and audience presence. Behind the surface, 559 LEDs are arranged on reflective aluminum-backed fabric to create diffused, shifting light. The front panel is made from bioplastic sheets using water, glycerin, starch, and gelatin, with some sections reinforced by muslin, leaf tannin heat-transfer patterns, and liquid dye. These patch-like units are joined together into a larger membrane and framed with wood sticks.

Together, the material surface and responsive light create a window that does not show an outside view, but produces the feeling of one. It extends the environment by giving the room a new atmospheric behavior: a wall that glows, senses, and remembers the presence of light.

Project logic

If light under 300, then led strips on

If light above 300, then led strips all off

If ultrasonic sensor read < 40 cm, color becomes warm and brighter

If ultrasonic sensor read > 40 cm, color is dimmer and colder

Drafts

Frame 10

each panel have few LED under, can space out individualy can light togehter like bent glass

window used to look outside, nostalgic, metaphorical window of mind?

more sound detected faster the blinking





1. Water (60mL)
2. Gelatin (6g)
3. Glycerin (6g, (3g makes a stiffer sheet)
4. Potato Starch (6g)



foil behind led to reflect light for more eveness

also considering to have distance sensor, when movements clos-

Starch Gelatin

This is a flexible sheet, potato starch+gelatin recipe that makes one 6-inch x 9-inch sheet (the recipe can be scaled up – we've made 4 feet x 4 feet sheets with these ratios)

Ingredients

1. Water (60mL)
2. Gelatin (6g)
3. Glycerin (6g, (3g makes a stiffer sheet)
4. Potato Starch (6g)
5. Shallow tray for drying (corrugated plastic works very well, which can be cut to the desired size with 1-inch edges folded up to create a tray)

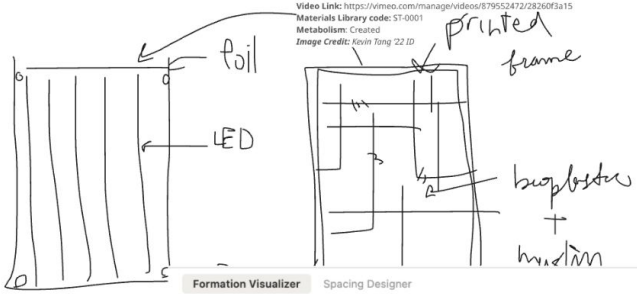
Process

1. Combine dry ingredients in a beaker
2. Add glycerin and water to the beaker.
3. Mix all ingredients well in a small beaker until incorporated.
4. Place on the hot plate and stir. Continue stirring until mixture begins to turn whitish and thickens slightly, like a mucus consistency
5. Carefully pour the liquid out into your tray
6. Quickly tilt the tray to make sure the liquid reaches all corners; place it on a heat-proof surface

Notes

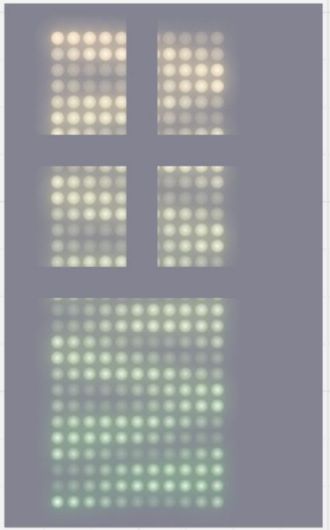
- Overheating or rapid stirring will cause bubbles to form that will impact your results
- Depending on temperature and humidity, bioplastic will take 4-7 days to cure before you peel it off the sheet. Don't let it sit longer than a week before peeling it off the tray
- The sheet needs to be slightly rough to constrain the bioplastic – too smooth (don't use glass or non-stick!) and the bioplastic will slip off of it and deform; too harsh and it will be hard to remove
- This recipe can be used as a base to create composites: consider adding dyes, fibers, paper scraps, etc.

Video Link: https://vimeo.com/manage/videos/879552472/28260f3a15
Materials Library code: ST-0001
Metabolism: Created
Image Credit: *Kevin Tang '22 ID*



Formation Visualizer Spacing Designer

less about object more about space



Bioplastic experimentation

Regular bioplastic with dye

And bioplastic on muslin





Other materials used:

Wood, yarn.

Trimming & Assembling

